

Quasi Newton Methods

min: $f(x)$

$$\text{GD} : x_{t+1} = x_t - \eta \cdot I \cdot \nabla f(x_t)$$

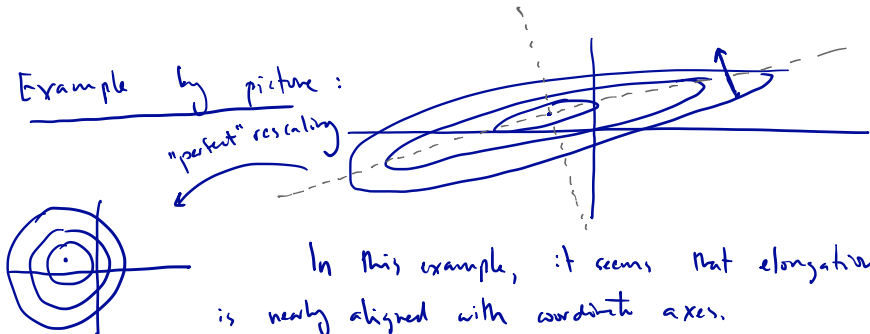
$$\text{Newton} : x_{t+1} = x_t - \eta \cdot (\nabla^2 f(x_t))^{-1} \cdot \nabla f(x_t)$$

$$\text{Quasi-N} : x_{t+1} = x_t - \eta \cdot H_t \cdot \nabla f(x_t)$$

Hope for : ① H_t more useful than I

② H_t is less expensive to compute than $(\nabla^2 f(x_t))^{-1}$

Example by picture :



In this example, it seems that elongation is nearly aligned with coordinate axes.

⇒ Might get much of speedup of Newton, by rescaling axes.

If we are only rescaling the axes -

$$y = Ax, \quad A \text{ } n \times n \text{ symmetric matrix,}$$

Only rescaling axes: $A = \begin{pmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix}$

For our update:

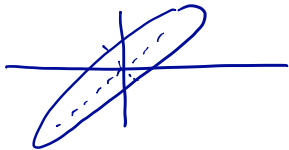
$$x_{t+1} = x_t - \eta H_t \nabla f(x_t)$$

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & & 0 \\ & \ddots & \\ 0 & & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \sim \text{diagonal} = \begin{pmatrix} \frac{1}{\left(\frac{\partial^2 f}{\partial x_1^2}\right)} & & \\ & \ddots & \\ & & \frac{1}{\left(\frac{\partial^2 f}{\partial x_n^2}\right)} \end{pmatrix}$$

$\uparrow$
linear ~~block~~

What if elongation is not well aligned w/ axes?

Picture:



Ex: $\nabla^2 f = \begin{pmatrix} 1 & 0.99 \\ 0.99 & 1 \end{pmatrix}$ (check: $k \approx 200$)

$$H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = I \quad \rightarrow \text{just GD} \dots$$